

**HOPEN, Svalbard and Jan Mayen**WMO: **010620**Lat: **76.509N**Lon: **25.013E**Elev: **33**StdP: **14.68**Time Zone: **1.00 (EUC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 3             | -16.7      | -12.9 | -20.9                         | 1.7 | -16.4 | -16.6 | 2.2 | -12.7 | 33.8                  | 15.8 | 30.3 | 18.2 | 12.7                  | 40   | 1.006 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 3.9                    | 45.1            | 43.6 | 43.8 | 42.5 | 42.7 | 41.5 | 44.1                | 44.8 | 42.9 | 43.6 | 41.7 | 42.5 | 14.6                 | 220  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 43.4                            | 41.6 | 44.6 | 42.1 | 39.5 | 43.3 | 40.9 | 37.8 | 42.2 | 17.2          | 45.0 | 16.6 | 43.7 | 16.1 | 42.6 | 52.5              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 30.1          | 26.9 | 24.2 | DB                         | -13.6 | 49.6               | 9.8 | 3.8                                                | -20.6 | 52.3       | -26.3 | 54.6       | -31.8 | 56.7       | -38.9 | 59.5 | (4) |
| (5)               |      |      | WB                         | -14.0 | 47.3               | 9.6 | 2.5                                                | -20.9 | 49.1       | -26.5 | 50.5       | -31.8 | 51.9       | -38.8 | 53.7 | (5) |

**Monthly Climatic Design Conditions**

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct  | Nov  | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)  | (o)  | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 25.9   | 17.0  | 15.9  | 13.5  | 17.5  | 26.4  | 33.2  | 37.7  | 38.7  | 36.3  | 30.1 | 24.2 | 19.6  |
|                                                                                  | DBStd       | 12.60  | 13.93 | 13.40 | 12.54 | 10.38 | 6.19  | 3.14  | 3.05  | 2.81  | 3.69  | 6.12 | 9.10 | 12.00 |
|                                                                                  | HDD50       | 8790   | 1022  | 956   | 1130  | 976   | 732   | 503   | 380   | 349   | 410   | 617  | 773  | 942   |
|                                                                                  | HDD65       | 14265  | 1487  | 1376  | 1595  | 1426  | 1197  | 953   | 845   | 814   | 860   | 1082 | 1223 | 1407  |
|                                                                                  | CDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | 0    | 0     |
|                                                                                  | CDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | 0    | 0     |
|                                                                                  | CDH74       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | 0    | 0     |
|                                                                                  | CDH80       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | 0    | 0     |
| Wind                                                                             | WSAvg       | 12.3   | 13.9  | 13.9  | 13.0  | 11.8  | 10.9  | 9.9   | 11.4  | 10.7  | 11.9  | 13.1 | 13.8 | 13.7  |
| Precipitation                                                                    | PrecAvg     | 17.0   | 1.4   | 1.5   | 1.4   | 1.1   | 0.8   | 1.1   | 1.3   | 1.6   | 1.8   | 1.8  | 1.7  | 1.5   |
|                                                                                  | PrecMax     | 29.6   | 3.1   | 5.3   | 2.9   | 2.5   | 2.8   | 2.6   | 4.0   | 3.9   | 3.0   | 3.5  | 3.9  | 3.9   |
|                                                                                  | PrecMin     | 12.4   | 0.4   | 0.2   | 0.3   | 0.3   | 0.2   | 0.2   | 0.2   | 0.5   | 0.5   | 0.8  | 0.5  | 0.6   |
|                                                                                  | PrecStd     | 3.6    | 0.7   | 0.9   | 0.7   | 0.6   | 0.6   | 0.6   | 0.8   | 0.8   | 0.6   | 0.7  | 0.8  | 0.8   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 36.9  | 36.7  | 35.8  | 36.7  | 38.2  | 43.0  | 47.3  | 47.8  | 45.9  | 43.2 | 39.2 | 37.7  |
|                                                                                  |             | MCWB   | 35.4  | 35.2  | 34.4  | 35.1  | 36.6  | 41.0  | 45.4  | 46.2  | 44.2  | 41.8 | 37.4 | 35.9  |
|                                                                                  | 2%          | DB     | 35.1  | 34.7  | 33.8  | 34.6  | 36.0  | 40.6  | 44.8  | 45.3  | 43.9  | 41.4 | 36.8 | 35.7  |
|                                                                                  |             | MCWB   | 33.9  | 33.5  | 32.7  | 33.6  | 34.6  | 38.7  | 43.3  | 44.0  | 42.8  | 40.2 | 35.6 | 34.4  |
|                                                                                  | 5%          | DB     | 33.0  | 32.3  | 31.8  | 33.1  | 34.5  | 39.1  | 43.3  | 43.9  | 42.7  | 39.4 | 35.1 | 33.8  |
|                                                                                  |             | MCWB   | 31.9  | 31.2  | 31.0  | 32.2  | 33.2  | 37.4  | 42.0  | 42.6  | 41.8  | 38.3 | 34.1 | 32.6  |
|                                                                                  | 10%         | DB     | 31.1  | 30.7  | 30.4  | 31.4  | 33.4  | 37.7  | 42.1  | 42.7  | 41.3  | 37.5 | 33.6 | 32.1  |
|                                                                                  |             | MCWB   | 30.0  | 29.8  | 29.8  | 30.6  | 32.2  | 36.2  | 40.8  | 41.5  | 40.3  | 36.6 | 32.7 | 31.1  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 35.9  | 35.8  | 34.9  | 35.6  | 36.9  | 41.4  | 45.8  | 46.7  | 44.8  | 42.3 | 37.9 | 36.5  |
|                                                                                  |             | MCDB   | 36.6  | 36.6  | 35.6  | 36.4  | 37.6  | 42.6  | 46.9  | 47.7  | 45.5  | 42.8 | 38.8 | 37.4  |
|                                                                                  | 2%          | WB     | 34.2  | 33.8  | 32.9  | 33.9  | 35.0  | 39.3  | 43.7  | 44.4  | 43.1  | 40.5 | 35.9 | 34.8  |
|                                                                                  |             | MCDB   | 35.1  | 34.8  | 33.6  | 34.4  | 35.9  | 40.4  | 44.5  | 45.1  | 43.8  | 41.5 | 36.7 | 35.6  |
|                                                                                  | 5%          | WB     | 32.0  | 31.3  | 31.2  | 32.4  | 33.6  | 37.8  | 42.3  | 42.9  | 41.9  | 38.6 | 34.3 | 32.9  |
|                                                                                  |             | MCDB   | 32.8  | 32.1  | 31.7  | 33.0  | 34.3  | 38.8  | 43.1  | 43.6  | 42.6  | 39.4 | 35.0 | 33.6  |
|                                                                                  | 10%         | WB     | 30.2  | 29.9  | 29.8  | 30.7  | 32.5  | 36.5  | 41.1  | 41.7  | 40.5  | 36.8 | 32.9 | 31.3  |
|                                                                                  |             | MCDB   | 30.8  | 30.6  | 30.3  | 31.4  | 33.3  | 37.6  | 42.0  | 42.4  | 41.1  | 37.5 | 33.5 | 32.0  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 7.7   | 8.3   | 8.5   | 7.9   | 5.4   | 4.1   | 4.5   | 3.9   | 3.6   | 4.4  | 6.0  | 7.3   |
|                                                                                  |             | MCDBR  | 7.2   | 7.4   | 8.3   | 7.2   | 5.7   | 5.9   | 6.2   | 5.7   | 5.4   | 5.6  | 6.2  | 7.5   |
|                                                                                  |             | MCWBR  | 7.1   | 7.4   | 8.4   | 7.2   | 5.3   | 5.0   | 5.4   | 5.0   | 5.1   | 5.9  | 6.2  | 7.5   |
|                                                                                  | 5% WB       | MCDBR  | 7.0   | 7.5   | 8.4   | 7.2   | 5.6   | 5.5   | 5.9   | 5.6   | 5.2   | 5.6  | 6.2  | 7.3   |
|                                                                                  |             | MCWBR  | 7.2   | 7.6   | 8.7   | 7.2   | 5.5   | 4.9   | 5.4   | 5.1   | 5.1   | 6.0  | 6.4  | 7.4   |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | N/A    | 0.226 | 0.275 | 0.304 | 0.321 | 0.307 | 0.318 | 0.311 | 0.284 | 0.356 | N/A  | N/A  |       |
|                                                                                  | taud        | N/A    | 1.899 | 2.060 | 2.170 | 2.286 | 2.455 | 2.547 | 2.544 | 2.454 | 2.646 | N/A  | N/A  |       |
|                                                                                  | Ebn at Noon | 0      | 28    | 179   | 239   | 259   | 272   | 264   | 246   | 197   | 42    | 0    | 0    |       |
|                                                                                  | Edh at Noon | 0      | 4     | 19    | 28    | 30    | 27    | 23    | 20    | 14    | 4     | 0    | 0    |       |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0      | 15    | 261   | 838   | 1228  | 1538  | 1182  | 746   | 286   | 42    | 0    | 0    |       |
|                                                                                  | RadStd      | 0      | 2     | 49    | 148   | 189   | 222   | 119   | 83    | 28    | 6     | 0    | 0    |       |

**Historical Trends**

|                             |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only           |  | +2.81 | +8.33   | +7.20  | N/A     | N/A   | N/A   | -970        | -968  | N/A   | N/A   |
| (47) Regional (0 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page